$i^{11} = -i$

$i^{24} = 1$

$i^{34} = i$

$(3i)^2 = 6i^2 = -6$

5. $\frac{\pi}{6} = 30^0$

6. $\frac{2\pi}{3} = 60^0$

7. $\frac{3\pi}{4} = 135^0$

8. $\frac{5\pi}{3} = 300^0$

9. $\frac{7\pi}{6} = 210^0$

10. $\frac{5\pi}{4} = 225^0$

11. $\pi + \frac{\pi}{4} = \frac{5\pi}{4}$

12. $\pi + \frac{\pi}{6} = \frac{7\pi}{6}$

13. $2\pi - \frac{\pi}{6} = -\frac{11\pi}{6}$

14. $2\pi - \frac{\pi}{6} = -\frac{11\pi}{6}$

15. $\frac{\pi}{2} + \frac{\pi}{6} = \frac{3\pi}{6}$

16. $\frac{3\pi}{4} - \frac{\pi}{2} = \frac{\pi}{4}$

17. $\frac{2\pi}{3} = 120^{\circ}$

2 четверть

$\cos = -\frac{1}{2} \quad \sin = \frac{\sqrt{3}}{2}$

18. $\frac{3\pi}{4} = 135^{\circ}$

2 четверть

$\cos = -\frac{\sqrt{2}}{2} \quad \sin = \frac{\sqrt{2}}{2}$

19. $\frac{5\pi}{4} = 225$

3 четверть

опорный угол $= -45$

$-45 = -\frac{\pi}{4}$

$\cos = -\frac{\sqrt{2}}{2} \quad \sin = -\frac{\sqrt{2}}{2}$

20. $\frac{5\pi}{3} = 300^\circ$

4 четверть

опорный угол $= 60^\circ$

$60^\circ = \frac{\pi}{3}$

$\cos = \frac{1}{2} \qquad \sin = -\frac{\sqrt{3}}{2}$

21. $\frac{7\pi}{6} = 210$

это 3 четверть

опорный угол $240 - 210 = 30^\circ$

$\cos = -\frac{\sqrt{3}}{2} \qquad \sin = -\frac{1}{2}$

22. $\frac{7\pi}{4} = 315$

4 четверть

опорный угол $= 45^\circ$

$45^\circ = \frac{\pi}{3}$

$\cos = \frac{\sqrt{2}}{2} \qquad \sin = -\frac{\sqrt{2}}{2}$

23. $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$\cos\frac{\pi}{3} = 60^\circ = \frac{\pi}{4} = \frac{\sqrt{2}}{2}$

$\sin\frac{\pi}{3} = \frac{\sqrt{2}}{2}$

$$z = 2\left(\frac{\sqrt{2}}{2} + i\,\frac{\sqrt{2}}{2}\right)$$

$$z = 2\cdot\frac{\sqrt{2}}{2} + 2\cdot\frac{\sqrt{2}}{2}i$$

$$z = \sqrt{2} + \sqrt{2}i$$

24. $z = 4\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$

$$\cos\frac{3\pi}{4} = \frac{\pi}{4} = -\frac{\sqrt{2}}{2}$$

$$\sin\frac{3\pi}{4} = \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

$$z = 4\cdot\left(-\frac{\sqrt{2}}{2}\right) + 4\cdot\frac{\sqrt{2}}{2}i$$

$$z = -2\sqrt{2} + 2\sqrt{2}i$$

25. $z = 2\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$

$$\cos\frac{4\pi}{3} = 240^\circ = 1$$

$$\cos\frac{4\pi}{3} = 0$$

$$z = 2 + i$$

26. $z = 6\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$

$\cos\frac{5\pi}{6} = 30^\circ = \frac{\pi}{6} = \frac{\sqrt{3}}{2}$

$\sin\frac{5\pi}{6} = \frac{1}{2}$

$z = 6 \cdot \frac{\sqrt{3}}{2} + 6 \cdot \frac{1}{2} i$

$z = 3\sqrt{3} + 12i$

28. $z = 2e^{i\,7\pi/2}$

$z = 2\left(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\right)$

$\cos\frac{7\pi}{4} = \frac{\sqrt{2}}{2}$

$\sin\frac{7\pi}{4} = -\frac{\sqrt{2}}{2}$

$z = 2 \cdot \frac{\sqrt{2}}{2} + 2 \cdot -\frac{\sqrt{2}}{2} i$

$z = \sqrt{2} - \sqrt{2} i$

27. $z = 3e^{i\pi/2}$

$z = 3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

$\cos\frac{\pi}{2} = 90^\circ = 0$

$\sin\frac{\pi}{2} = -1$

$z = -3i$

29. $z = \sqrt{3} + i$

$r = \sqrt{3^2 + 1^2} = \sqrt{4} = 2$

$\cos\varphi = \frac{\sqrt{3}}{2} = \frac{\pi}{6}$

$z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

30. $z = 1 + \sqrt{3}i$

$r = \sqrt{1^2 + \sqrt{3}^2} = \sqrt{1+3} = \sqrt{4}$

$\cos\varphi = \frac{1}{2} = \frac{\pi}{3}$

$z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

31.

$$z = -\sqrt{3} + i$$

$$r = \sqrt{-\sqrt{3}^2 + 1^2} = \sqrt{3+1} = 2$$

$$\cos\varphi = \frac{-\sqrt{3}}{2} = -\frac{\sqrt{3}}{2} = \frac{\pi}{6}$$

$$z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

32. $z = -1 - \sqrt{3}i$

$$r = \sqrt{-1^2 - \sqrt{3}^2} = \sqrt{1+3} = 2$$

$$\cos\beta = -\frac{1}{2} = \frac{\pi}{3}$$

$$z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

33. $z = -\sqrt{2} - \sqrt{2}i$

$$r = \sqrt{-\sqrt{2}^2 - \sqrt{2}^2} = \sqrt{2+2} = 2$$

$$\cos = \frac{-\sqrt{2}}{2} = \frac{\pi}{4}$$

$$z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

34. $z = 3i$

$r = \sqrt{0^2 + 3^2} = \sqrt{9} = 3$

$\sin = \frac{3}{3} = 1 = \frac{\pi}{2}$

$z = 3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

35. $z^2 - 6z + 13 = 0$

$a = 1 \quad b = -6 \quad c = 13$

$D = -6^2 - 4 \cdot 1 \cdot 13$

$-36 - 52$

$D = -88$

$z_{1,2} = \frac{-6 \pm \sqrt{88}i}{2} = -3 \pm \sqrt{88}i$

36. $z^2 + 2z + 5 = 0$

$a = 1 \quad b = 2 \quad c = 5$

$D = 4 - 4 \cdot 5$

$D = -16$

$\sqrt{D} = 4i$

$$z_{1,2} = \frac{-2 \pm 4i}{2} = -1 + 4i;\ 1 - 4i$$

37. $z^2 + 8z + 25 = 0$

$a = 1 \quad b = 8 \quad c = 25$

$D = 81 - 4 \cdot 25$

$D = -19$

$$z_{1,2} = \frac{-8 \pm \sqrt{19}\,i}{2} = -4 + \sqrt{19}\,i;\ -4 - \sqrt{19}\,i$$

38. $z^2 - 4z + 8 = 0$

$a = 1 \quad b = -4 \quad c = 8$

$D = 16 - 4 \cdot 8$

$D = -16$

$\sqrt{D} = 4i$

$$z_{1,2} = \frac{4 \pm 4i}{2} = 2 + 4i;\ 2 - 4i$$

39. $2z^2 + 4z + 10 = 0$

$a = 2 \quad b = 4 \quad c = 10$

$D = 16 - 40$

$D = \sqrt{-24}$

$z_{1,2} = \dfrac{-4 \pm \sqrt{24}i}{4} = -1 + \sqrt{24}i;$

$-1 - \sqrt{24}i;$

40. $z^2 - (3+i)z + 3i = 0$

$a = 2 \quad b = -(3+i) \quad c = 2 + 3i$

$D = -(3+i)^2 - 4 \cdot 2 \cdot 2 + 3i$

$9 + i^2 - 8(2 + 3i) = 9 + i^2 - 16 + 3i =$

$-7 + 4i$

$z_{1,2} = \dfrac{(3+i) \pm -7 + 4i}{4} = \dfrac{10 + 5i}{4} =$

$\dfrac{5}{2}$

$z^2 = -9$